CHAPTER

27

Geriatric Restorations

The relationship of oral health/diseases with the systemic health, particularly in the elderly, is well established. Therefore, retention of teeth can have an impressive impact on the overall physical and mental health of an individual.

The term geriatrics stems from a Greek word 'Geron' meaning 'old man' and 'Iatros' meaning 'healer'. The chronological age of 'old' is dynamic and keep changing with time. The groups recognized by consensus are: Young old (65–74); older old (75–84) and the old (more than 85). Geriatric dentistry is defined as "*the delivery of dental care to older adults involving diagnosis, prevention and treatment of problems associated with normal aging and age-related diseases as part of an interdisciplinary team with other health care professionals*".

Life expectancy has doubled in the past century, more so in the last two decades. This improved life expectancy is a result of improved health awareness, accessibility, affordability and acceptability of improved medical care.

Old age is associated with several risk factors; general as well as specific to the oral cavity. Managing compromised dentition is indeed a multi-faceted challenge. The general risk factors include various intercurrent and psychological problems. The risk factors related specifically to oral cavity are gingival recession and age-related odontometric changes.

The health care providers should be familiar with the course and the complications associated with the systemic conditions. For example, patients with some form of cardiovascular disease are vulnerable to physical/emotional stress, which may be encountered during dental treatment.

Some patients may not be able to tolerate reclining position on the chair. The elderly are more susceptible to orthostatic hypertension, hence need special attention, especially when they move from a reclining posture to a standing position. Vasoconstrictors should not be administered to patients with unstable angina, uncontrolled hypertension or people undergone coronary bypass surgery. Prophylactic antibiotics may be necessary for patients with history of high-risk cardiac conditions, while undertaking surgical procedures; however, restorations can be placed without prior medication. The psychoemotional chord of the elderly patients should always be respected and taken care of; individuals be motivated as regard their life expectancy and prognosis of the treatment.

The elderly individuals may suffer from decline in their cognitive functions. Senile dementia, a common phenomenon amongst the elderly, may lead to confusion, difficulty in making decisions, comprehension and even alter the patients' ability to learn and concentrate on new tasks.

AGE CHANGES IN DENTAL TISSUES

Aging of an individual is manifested at all levels of life. Age changes are considered to be the result of day-to-day biological activities. Age changes are slow or non-existent in tissues that do not exhibit biological reactivity. It is often difficult to differentiate between physiological and the pathological age changes. Aging can be defined as the combination of process beyond development and maturation, which results in a diminution of capacity of the tissues. The physical state of aging is referred to as '*senescence*' and the study of the aging process is known as '*gerontology*.'

Effects of Aging

The general features encountered in aging are:

- Tissue desiccation
- Reduced elasticity
- Diminished reparative capacity
- Altered cell permeability
- Keratinization is diminished (thin dermis and epidermis)
- Decreased blood supply
- Degeneration of nerve endings
- Capillaries are fragile; may result in development of hematomas even after a minor trauma

- Bone undergoes osteoporosis; trabeculae are reduced in number, the cortical plates are thinned, vascularity is reduced, lacunae resorption is more prominent, and susceptibility to fracture is increased.

MECHANISM OF AGING

The mechanism of aging is categorized into four types:

i. Biologic clock
ii. Immunological
iii. DNA damage related
iv. Damage of other cellular components

The mechanism of aging has been summarized in following two theories:

a. Cross-Linkage Theory

According to this theory, aging starts with the formation of cross-linkages in proteins and nucleic acids. Cross-linkages can also occur between molecules of collagen, which makes it tougher and more rigid. Cross-linked collagen is more readily infiltrated with calcium deposits. A previous aging theory, known as '*calcium theory*', linked aging to the gradual accumulation of calcium deposits throughout the body. Once a substance is cross-linked, it eventually loses its function. It is considered that cross-linkage prevents certain protein and nucleic acid molecules from being metabolized by the body's enzymes, which then accumulate in the cells.

Aging leads to accumulation of cross-linkage agents. These agents can be derived from foods, drugs, and pollutants.

The cross-linkage agents include acetaldehyde, formaldehyde, pyruvic acid, citric acid, lipid peroxides, silicon, lead, aluminum, copper, iron, calcium, zinc, etc. Glucose can attach to proteins and nucleic acids to form cross-linkage products. Tobacco smoke contains acrolein and glycerin, which are considered to be cross-linkage agents. Cross-linkages can induce somatic mutations and protein degeneration; subsequently, change the immunologic behavior.

b. Free-Radical Theory

Free radicals can propagate through hundreds of thousands of molecules and wreak havoc before they encounter another free radical and become neutralized. Free radicals have been called the "*great white sharks of the biochemical sea of life*". If unchecked, free radicals can continue to create other free radicals from normal molecules, thereby disrupting vital functions. In this way, free radicals can cause serious effects to cellular metabolism as a consequence of damage to DNA, RNA, enzymes, lipids, immune cells, and cell membranes. Free radicals are produced as byproducts of normal oxygen metabolism and oxidation of unsaturated fatty acids.

This theory states that oxygen-derived free radicals cause lipid oxidation, which in turn damages cell membranes and other cellular structures. Mitochondria are the main source of free radicals. The body naturally contains antioxidant mechanisms that quench free radicals. With aging, these mechanisms become less effective and more free radicals are generated, leading to tissue damage.

AGING AND DENTAL TISSUES

Aging affects all tissues of oral cavity; described as follows:

I. Age Changes in Enamel

a. Wearing of the enamel (loss of vertical dimension and flattening of proximal contours).
b. The surfaces of recently erupted teeth are covered with pronounced rod ends and perikymata. As the age advances, there is generalized loss of rod ends and flattening of the perikymata. Rate at which the structure is lost depends on location of the surface of the tooth and on location of the tooth in the oral cavity.
c. Pronounced wear facets.
d. Surface darkening (Localized increase of nitrogen and fluoride and change in organic content lead to darkening of enamel surface).
e. Enamel becomes less permeable.

II. Age Changes in Dentin

a. ***Secondary dentin:*** The major age change in dentin is the formation of secondary dentin. Secondary dentin is deposited only after completion of primary dentin.

A few authors describe secondary dentin as regular and irregular. The regular secondary dentin is considered physiological; whereas, irregular secondary dentin is considered as pathological. The irregular secondary dentin formed as a result of abnormal irritation/stimulation is also called '*reactionary dentin*' or '*tertiary dentin*'. Reactionary dentin differs from reparative dentin as reactionary dentin is generalized deposition by the pre-existing odontoblasts and reparative dentin is localized deposition formed by newly differentiated odontoblasts in response to an aggressive local stimulus.

This process of secondary dentin formation results in gradual reduction in size of the pulp chamber and root canals. The number of dentinal tubules is

reduced since irregular secondary dentin obliterates tubule openings. The progress of caries, attrition, abrasion, erosion or cavity preparation generates sufficient stimuli causing deposition of collagen fibres and apatite crystals in the dentinal tubules.

b. ***Dentinal sclerosis:*** The exact mechanism of dental sclerosis or the deposition of calcium salts in the tubules is not properly understood; the likely source of calcium salts is within the dentinal tubules. The increased mineralization of the tooth decreases the conductivity of the odontoblastic processes. Apatite crystals are initially sporadic in dentinal tubules, but gradually the tubules get filled with a fine meshwork of crystals. Sclerosis reduces permeability of dentin and may help prolong pulp vitality. Transparent or sclerotic dentin can be observed in teeth of elderly people, especially in roots. It appears transparent or light in transmitted light and dark in reflected light due to the equalization of refractive indices of sclerosed dentinal components.

c. ***Dead tracts:*** When caries progresses rapidly, the odontoblasts degenerate. The dentinal tubules are no longer filled with living protoplasm, resulting in dead tracts. The remaining vital odontoblasts and other pulp cells attempt to seal off the dead tracts by elaboration of the reparative dentin.
Dead tracts are seen in ground sections of teeth and are manifested as a black zone by transmitted light and as a white zone in reflected light. This optical phenomenon is due to differences in the refractive indices of affected tubules and the normal tubules. The nature of change in the affected tubules is not known, although these tubules are not calcified and are permeable to the penetration of dyes.

d. ***Tertiary dentin:*** Tertiary dentin develops when pulp irritants are more intense. Tertiary dentin differs from secondary dentin, being localized adjacent to the irritated zone, tubules irregular, tortuous, and reduced in number or absent.

III. Age Changes in Pulp

a. *Reduction in Size and Volume of Pulp*

- The reduction in coronal pulp is the result of continual apposition of dentin occlusally as well at furcation area (Deposition of dentin at furcation area is greater than that of occlusal dentin).
- Secondary dentin formation continues throughout life and may result in complete pulpal obliteration (Fig. 27.1A, B). In maxillary anterior teeth, the secondary dentin is formed on lingual of the pulp chamber; in molars, the maximum deposition takes place on floor of the chamber.

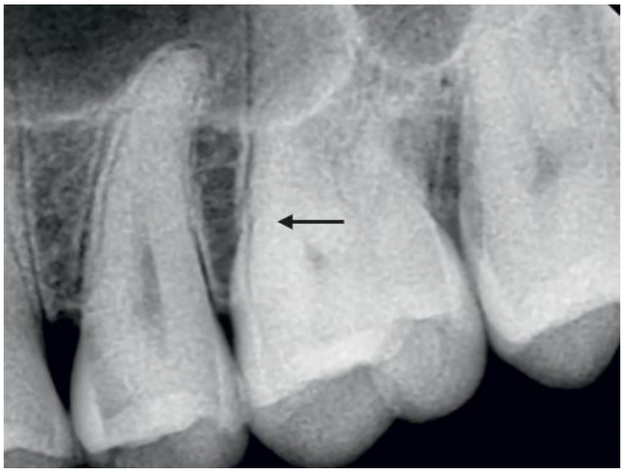


Fig. 27.1A: Pulpal obliteration in aged dental pulp (arrow)



Fig. 27.1B: Cut section of tooth showing reduced pulp space and sclerotic dentin

b. *Decrease in Number of Cells*

- Number of cells are reduced by 50% in aged pulps.
- Fibroblasts show degeneration characterized by small size and a decrease in number of organelles.
- Degeneration of odontoblasts.
- Increase in vacuole numbers and gradual degenerative changes leading to localized or generalized absence of cells.

c. *Presence of Calcifications*

The types of calcifications recognized in dental pulp are:

i. Discrete calcification
ii. Diffuse (dystrophic) calcification

i. ***Discrete calcification:*** Pulp stones or denticles are discrete calcifications found freely within the pulp tissue or attached/embedded in dentin. Denticles are further classified according to structure and location.

According to structure, the denticles are:

- *True denticles:* Denticles possessing a central cavity filled with epithelial remnants, surrounded peripherally by odontoblasts.
- *False denticles:* Compact degenerative masses of calcified tissues.

The differences between true and false denticles are summarized in Table 27.1

Table 27.1: Difference between true and false denticles

True denticles	*False denticles*
Attached to the wall of pulp chamber	Not attached to the wall of pulp chamber
Also known as attached denticles	Also known as free denticles
More common	Less common

According to location, the denticles are:

- *Free denticle:* Freely lying in pulp chamber
- *Attached denticle:* Attached to the dentinal walls
- *Interstitial denticle:* Surrounded by secondary denticle

Etiology: The incidence of pulp calcification increases with age without any definite association with pulpal inflammation. A few authors suggested that microbes are responsible for the pulp stones; others hypothesized thrombosis and/or vascular injury as possible mechanisms for pulp stone formation.

Characteristic features:

- A common feature above the age of 45 years.
- Mostly seen in root pulp as an isolated calcified mass; may coalesce to form large mass that fuses with dentin.
- Calcified masses seen in coronal pulp may fuse together to obliterate the pulp architecture.
- With increased cross-linkage of collagen during advancing age, there is a tendency for these fibers to get mineralized.

ii. ***Diffuse (dystrophic) calcification:*** Diffuse (dystrophic) calcification refers to the bio-mineralization of the pulp tissue. The calcification, generally a pulpal response to trauma, is characterized by deposition of hard tissue within the root canal space. This is commonly seen in young adults in anterior region of the oral cavity.

d. Decreased Vascular Supply

The small arteries in the root pulp show osteoclastic activity with age. The intima of the vessel gets thickened resulting in smaller lumen. The old blood capillaries exhibit a widening of the basement membranes; whereas, the young vessels appear as thin membrane. In the process of diffuse calcification of the root pulp, the adventitia and media of small arteries, veins and also walls of the capillaries get mineralized, eventually, leading to loss of vessels in the old pulps.

Capillary permeability also decreases with age. Transfer of substances from blood to cells may be depleted during aging.

e. Decreased Nerve Supply

The nerves within the root pulp exhibit changes as a result of diffuse calcifications. The connective tissue sheath of the nerve bundles becomes mineralized. The cuspal nerves are decreased and their terminal branching is lost. Persistent nerve fibres show signs of degeneration, such as, reticulation, fragmentation, beading, etc.

IV. Age Changes in Cementum

The process of depositon of cementum with age, termed as '*hypercementosis*', is an abnormal thickening of cementum. It may be diffuse or circumscribed. It can affect all the teeth; may confine to single tooth, or even affects only parts of one tooth. The thickness of young apical cementum is 100 to 200 µm, which may increase two or three times, as the age advances.

The causes of hypercementosis are:

- Accelerated elongation of tooth
- Inflammation around the tooth
- Tooth repair
- Paget's disease of bone

Accelerated elongation of tooth owing to loss of an antagonist, leads to hyperplasia of cementum (may be to maintain the normal width of the periodontal ligament). Inflammation at the apex of the tooth root stimulates excessive deposition of cementum. In such case, cementum is laid down on the root surface at some distance above the apex. Tooth repair does not, always, result in deposition of the cementum. Occlusal trauma may result in mild root resorption. Such root resorption is repaired by deposition of cementum between the root fragments as well as on their periphery.

V. Age Changes in Bone

The characteristic changes in bone with age are:

- Density of bone is decreased with age; more so in women (being less dense, the bone is more brittle).
- Lamina dura is often lost and the cortical bone becomes thinner, especially at the angle of the mandible.
- Porosity is increased, mainly due to increase in vascular spaces.

- Volume occupied by each canaliculus remains constant or may decrease.
- Lacunar volume is increased, despite a reduction in number of lacunae.
- Walls of blood capillaries get thickened.
- Active cells of bone are reduced in number.

VI. Age Changes in Periodontium

a. ***Changes in gingiva and other areas of oral mucosa***

- Diminished keratinization
- Reduced amount of stippling
- Increased width of attached gingiva
- Decreased connective tissue cellularity
- Increase in amount of intercellular substances
- Reduced oxygen consumption
- Increase or no change in mitotic index of gingival epithelium
- Thinning of oral epithelium
- Increased keratinization of lip and cheek mucosa
- Atrophy of connective tissue with loss of elasticity
- Decrease in number of protein bound elements and mucoproteins
- Increase in number of mast cells.
- Oral mucosa possesses self-cleaning mechanism provided by turnover of epithelial cells, which decreases with age (Mucosal epithelial cells synthesize substances that are critical for maintenance of mucosal surface, viz. keratin, heminin).

b. ***Changes in periodontal ligament***

- Increased number of elastic fibres
- Decrease in vascularity, mitotic activity, and the number of collagen fibers
- Decrease in mucopolysaccharides
- Increase in arteriosclerotic changes
- Reduction in width caused due to lower functional demand owing to the decrease in strength of the masticatory musculature
- Increase in width caused due to availability of fewer teeth to support the entire functional load
- Decreased width because of continuous deposition of cementum and bone.

VII. Age Changes in Salivary Glands

The diminished function of salivary glands is common in geriatric individuals. With advancing age, there is an atrophy of acinar tissue, coupled with degenerative changes in the salivary glands. Decreased salivary flow may not be directly related to the aging; however, intake of certain drugs during aging might affect salivary secretion. The diminished flow of saliva/absence of saliva has many deleterious consequences for the elderly individual.

VIII. Miscellaneous

- It is hypothesized that the older adults may exhibit adverse reaction to injury or they might be more resistant because of decreased permeability of dentin (no proven documentation).
- No change in periradicular tissues as regard cellularity and vascularity with age.
- Similar pattern of healing in young and old (critical to healing is vascularity).
- Systemic conditions may affect restorative treatment prognosis at any age.

CLINICAL IMPLICATIONS

The clinical implications of the effects of aging are as follows:

I. Restorative Aspects

a. ***Attrition:*** With age, the enamel surface wears off due to tooth-to-tooth contacts during various functional/parafunctional movements of the mandible. Attrition may result in exposure of dentin, loss of vertical dimension and flattening of proximal contour. Abnormal attrition may relate to habits, such as tooth grinding or bruxism. The attrited teeth are preserved and covered with full crowns.

b. ***Cavity design:*** With age, the dentin loses its resiliency (formation of transparent dentin); subsequently, chances of fracture due to occlusal stresses are increased. The loss of resiliency (reduced organic components) of dentin can increase the susceptibility of tooth to cracks and cuspal fractures. Appropriate cavity preparation without any overhanging and unsupported enamel should be preferred.

c. ***Bonding:*** Partial or complete obliteration of the dentinal tubules is commonly observed with age. Even after increasing the etching time, the sclerotic obliterations in the dentinal tubules cannot be completely removed, resulting in minimal or no resin tag formation. Further, the zone of resin-impregnated sclerotic dentin is thinner as compared to normal dentin. The tensile bond strength to sclerotic root dentin with some contemporary adhesives was found to be 20–45% lower than those bonded to normal dentin.

II. Pulp Testing

The pulp testing procedures should be slow and gentle in aged patients. The cracked teeth should always be considered when pulpal or periapical disease is observed. The reduced neural and vascular components of aged pulps, reduced pulp volume and

change in the ground substance lead to delay in response as compared to younger pulps. The electric pulp testers, carbon dioxide snow and dichlorodifluoromethane, were found to be more reliable than ethyl chloride or ice in producing a positive response. Thermal tests are considered more reliable than electric pulp tests.

It is emphasized that pulpal symptoms are chronic in older patients, therefore, other sources of orofacial pain should be ruled out when pain is not localized. Diffuse pain of vague origin is uncommon in older pulps and limits the need for selective anesthesia.

III. Radiographs

The techniques of taking radiographs do not differ much among older adults; however, physiologic and anatomic changes may significantly affect their interpretation. The presence of tori, exostoses, and denser bone may require increased exposure times for proper diagnostic contrast. Pulp recession is accelerated by reparative dentin deposition. Receding pulp horns that are apparent on a radiograph may remain microscopically much higher. Narrow canals are the rule in old patients. The narrowest point in the canal may be difficult to determine; it is positioned farther from the radiographic apex because of continued cementum deposition.

IV. Access Preparation and Working Length Determination

Identification of canal orifices is the most difficult aspect of providing root canal treatment to older patients. The decreased pulp chamber size makes the canal orifice location difficult. The presence of pulp stone creates problems in the biomechanical cleaning and shaping the root canals. Although aging may reduce the volume and coronal extent of the chamber or canal orifice, its buccolingual and mesiodistal portions remain the same. In addition, the cementodentinal junction (CDJ) moves farther from the radiographic apex with continued cementum deposition, leading to difficulty in accurate working length determination.

V. Tooth–Periodontium Relationship

Attrition is the common feature of aging. Occlusal wear reduces cusp height and resultant increase in food table area.

The process of attrition is related to changes, such as continuous tooth eruption and gingival recession. As the tooth erupts, cementum is usually deposited in the apical region of the root. The reduction in bone height that occurs with aging is not necessarily related to occlusal wear.

If bone support is reduced, the clinical crown tends to become disproportionately long and exerts excessive leverage on the bone. By reducing the clinical crown length, attrition appears to preserve the balance between tooth and its bony support.

Wear of teeth on the proximal surfaces is accompanied by mesial migration of the teeth. Proximal wear reduces the anteroposterior length of the dental arch by approximately 0.5 cm by the age of 40. Anteroposterior narrowing from proximal wear is greater in teeth that taper toward the cervical aspect, such as incisors. Progressive attrition and proximal wear result in a reduced maxillary–mandibular overjet in the molar area and on edge-to-edge bite anteriorly.

VI. Masticatory Efficiency

Slight atrophy of the musculature is considered as a physiologic feature of aging.

The reduced masticatory efficiency in aged individuals is due to:

- Unreplaced missing teeth
- Loose teeth
- Poorly fitting dentures
- Unwillingness to wear dentures

Reduced masticatory efficiency may lead to poor chewing habits and the associated digestive disturbances.

Avitaminosis is relatively common in aged individuals (An adequate intake of vitamin, calcium, iron and potassium is particularly important to geriatric patients).

VII. Root Caries

Root caries is one of the common findings in older adults. Both active (progressive) and inactive (non-progressive) root caries can be observed in aged individuals.

The associated features, which lead to development of root caries are:

i. *Saliva:* Aging per se has no significant clinical impact on salivary secretion. The salivary hypofunction in the elderly is mostly due to intake of medications.
ii. *Gingival recession:* Root surface caries cannot develop unless preceded by gingival recession (Fig. 27.2). With advancing age, the loss of periodontal attachment is accompanied by gradual recession of gingiva (Fig. 27.3).
iii. *Age and sex:* Root caries increases with age. Men are generally more affected than women.

TREATMENT PLANNING FOR ELDERLY

Treatment planning in elderly involves patient's perceived needs and priorities as well as their expectations.



Fig. 27.2: Root caries with gingival recession



Fig. 27.3: Gingival recession

The operator should concentrate on patient's quality of life within domains of functions, symptoms, ability/disability and esthetics.

The following determinants are important prior to treatment planning:

- Desires and expectations
- Patients' dental needs
- Impact on patient's quality of life, viz. ability to eat, comfort level and esthetics.
- Probability of positive treatment outcome (prognosis)
- Ability to tolerate treatment stress
- Ability to maintain oral health
- Patients financial resources (whether individual has income or dependent on others)
- Family support (living alone or with family members)

Four types of needs for any dental treatment have been hypothesized, viz. functional, psychological, perceived and normative. One should integrate these needs into a holistic approach to achieve patient's quality of life.

Geriatric patients can be grouped into three categories while planning treatment strategies, viz. i) functionally independent, ii) frail, iii) functionally dependent.

i. ***Care for functionally independent individuals:*** The treatment is planned in following stages:

Stage I—Emergency phase: Initially, emergency treatment is carried out to relieve pain, if any.

Stage II—Maintenance phase: It includes management of infection, root planing and curettage, restoration of carious lesions, and patient education to improve oral health (such as, changes in dietary habits, plaque control and use of fluorides).

Stage III—Rehabilitation phase: It includes restoration of occlusal plane and rehabilitation of vertical dimension with fixed prosthesis.

ii and iii. ***Care for frail and functionally dependent individuals:*** These patients require assistance even for their normal day-to-day activities. Such patients should be given minimum treatment required to maintain physical and psychological comfort. Physician's presence may be necessary in some patients. ART techniques may prove useful in these cases for restorative management. Shay (1994) suggested a systematic approach to plan oral care for the geriatric patients, termed as OSCAR (Table 27.2).

Table 27.2: OSCAR: Planning geriatric patient

O: Oral: Evaluates teeth, mucosa, periodontium, status of pulp, saliva and occlusion
S: Systemic: Evaluates normative age changes, medical problems, drugs intake and interdisciplinary communication
C: Capability: Evaluates functional ability, such as oral hygiene, need for mobility and transportation
A: Autonomy: Evaluates ability to give consent or dependent on others
R: Reality: Evaluates prioritization of oral health care, financial limitations, etc.

Physiological changes with age

- Behavior changes
- Diet intake may be less
- Oral and systemic health may deteriorate
- Skin and blood vessels lose elasticity
- Delayed healing
- Bones become brittle
- Alteration in taste/smell
- Change in tooth anatomy
- Variations in pulpal physiology
- Color changes
- Continuous thickening of cementum
- Gingiva becomes edematous and friable; loss of stippling
- Salivary flow decreases

Systemic Diseases Influencing Treatment Planning in Elderly

A thorough medical history should be recorded, analyzing the probability of any systemic disease. (Patient's physician should be consulted.)

The systemic conditions, which warrant extra-attention before any restorative treatment in older adults, are as follows:

a. *Cardiovscular Diseases*

In patient with cardiovascular disease, it is advisable to minimize any kind of stress in the clinic. Hypertensive and patient taking anticoagulants should be treated with utmost care. In any case where the patient is at risk, the treatment is preferred under antibiotic premedication.

Vasoconstrictors should not be administered to patients with unstable angina, recent myocardial infarction, recent stroke, recent arrhythmias and hypertensive patients. Local anesthetic agents without vasoconstrictor or minimal vasoconstrictors are preferred for non-surgical procedures.

Vasoconstrictors should also be minimized in patient's undergone coronary bypass surgery, especially in first three months. These patients may not require antibiotic prophylaxis unless there are other complications.

Patients with artificial heart valves are considered highly susceptible to bacterial endocarditis. Such patients should be given prophylactic antibiotics before initiating any treatment.

Key points

- Check whether hypertension is under control or not
- Carefully administer local anesthesia
  Even though adrenaline is contraindicated in hypertensives, amount available in 1:100000 solutions is safe. As anesthetics without vasoconstrictors have a less effect and shorter duration of action, slow injection (one minute for 1.8 mL) using 26 or 27 gauge needle is preferred.
- Aged patients under hypertensive drugs may suffer from orthostatic blood pressure; operator should elevate the back of chair slowly to the upright position.

b. *Pregnancy*

Unless emergency treatment is warranted, it is advisable to defer elective treatment, especially during the first trimester, because of potential vulnerability of the fetus. The second trimester is the safest period during which any procedure can be carried out. Treatment planning should be directed at eliminating pain symptoms during pregnancy.

The postpartum period, wherein the mother breast-feed the child, should also be considered. The clinician should avoid any drug harmful to the infant.

Patient's gynecologist should be consulted as regard treatment needs of the patient and also the drug regime.

Key points

- Proper chair position for the comfort of the patient.
- Organogenesis is taking place in first trimester; any medication given has to be cleared by the gynecologist. Second trimester is relatively safe. Third trimester is also safe but care has to taken with chair position as the fetus is growing and postural hypotension may develop.
- To prevent supine hypotensive syndrome in the dental chair, the pregnant woman should have the right hip elevated (10 to 12 cm) or placing the patient in a 5.0 to 15% tilt on her left side can relieve pressure on the inferior vena cava. If hypotension is still not relieved, a full left lateral position may be helpful.

c. *Oral Cancer*

Many a times, the periapical radiolucency can be due to malignant lesions of the jaws. Whenever, there is a discrepancy between initial diagnosis and the clinical findings, the patient should be referred for second opinion. Patient undergoing chemotherapy and/or radiation to the head and neck may have impaired healing responses.

To minimize risk of osteoradionecrosis, no treatment is advised two weeks before radiation therapy. Prophylactic use of antifungal drugs can help prevent candidiasis. Glass-ionomer is effective in restoring caries due to radiations.

d. *Renal Diseases (Dialysis)*

Patient with renal failure and other kidney dysfunction may not be able to metabolize certain drugs. Hemodialysis tends to aggravate bleeding tendencies through physical destruction of platelets. The patients, who have had hemodialysis session, generally have bleeding tendencies and are fatigued. The elective treatment should be postponed till a day after hemodialysis. The operator should also be aware of the fact that hemodialysis removes certain drugs from the circulating blood and can shorten the effect of prescribed medications. The drugs used, if any, may have to be avoided or their doses are to be adjusted. Penicillin V and cephalexin requires increase in dosage; however, erythromycin requires no adjustment. The patient's physician should be consulted as regard specific drug requirements.

e. *Diabetes*

The controlled diabetic patients, free from complications, like renal failure, hypertension and/or coronary atherosclerotic disease, can be treated in routine. The

standard protocol is followed, wherein prophylactic antibiotics are prescribed.

Key points

- Check whether diabetes is under control or not (stage of diabetes).
- Patients with diabetes are prone to infection and delayed healing; hence, necessary precautions should be taken.
- Schedule early morning appointments, where the patient has had medication and meals, so that the risk of hypoglycemia is minimized.

f. *Blood Dyscrasia*

The systemic disorders are not definite contra-indications for executing any dental procedure; however, the operator should be conscious of the implications of these diseases.

- Check bleeding and clotting time
- Less traumatic procedure should be carried out (Most challenging is to treat a patient with hemophilia as there is always the risk of puncturing blood vessel and inducing internal bleeding particularly with inferior alveolar nerve blocks. If vital pulp extirpation has to be carried out, gradual fixation of pulp tissue followed by removal in multiple visits is the preferred choice. Care should be exercised in placing rubber dam clamps so that they do not impinge on gingival tissue. Notches may be placed on the labial and lingual surfaces into which the clamp's prongs will fit).

g. *Alzheimer's Disease/Parkinson's Disease*

Patients with Alzheimer disease/Parkinson's disease (neurodegenerative disorders) may present with concomitant hyposalivation (xerostomia) reducing buffering capacity, leading to dry and chapped lips, gingival bleeding, caries and other related problems.

- In the early stages, the treatment can be provided without any modification; in mid stages, pain is kept minimum while performing any procedure.
- Orthostatic blood pressure should also be looked into and manage accordingly.
- The sharp instruments should be used with utmost care because of patient's involuntary movements.

h. *Asthma*

Asthma, hay fever, skin rashes, etc. indicate hypersensitivity problems and the patients should be managed accordingly.

- The patient's medication and nebulizer should be kept ready.
- Clinic set up should be aerosol free.
- Patients may need physician clearance and medication even in routine procedures.

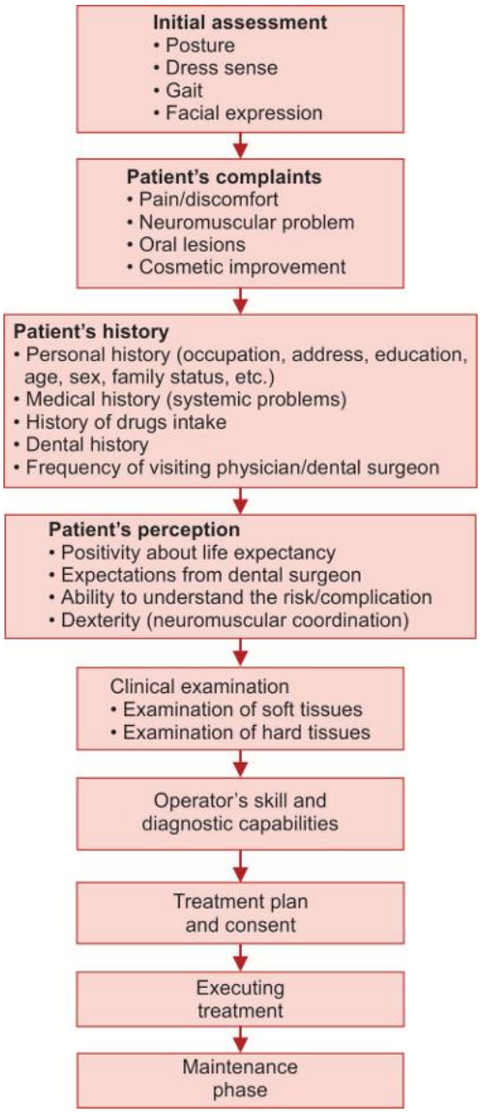
i. *Medications*

Following additional features, as regards drug intake, also influence treatment planning.

- Whether patient is on medication: if yes, for which problem?
- Whether the medication was prescribed by any qualified doctor? This helps in identifying patient's attitude towards his/her ailments (Some medication may be interfering with the antibiotics and analgesics; need to be analyzed).
- Allergic to any drug.

The treatment planning in elderly patients is summarized in Flowchart 27.1

Flowchart 27.1: Treatment planning in elderly



MANAGEMENT OF COMMON ORAL DISEASES IN ELDERLY

The following features are important for management of common oral diseases in elderly patients.

- Educating the need of oral health on periodic basis. Principle of Tell-Show-Do ('Tell', to explain; 'show' to demonstrate and 'do' to guide the individual to practice) be communicated to the elderly.
- For patients with impaired cognitive function, due to stroke, Alzheimer disease, etc., a family member or a professional caregiver should be educated as regard daily oral health needs.
- Frequent recall with fluoride application, especially for individuals at high risk of caries; also for those who are unable to maintain oral hygiene.
- For individuals with physical disability or limited dexterity, improved techniques of brushing/flossing (electric brush, water flossing) can be suggested.
- For individuals with dry mouth, agents which relieve dry mouth can be advised (sugar-free lozenges or xylol-containing lozenges, alcohol-free mouthrinses, gums, etc.). Encourage them to hydrate with water frequently.
- Periodic monitoring by dental professional is mandatory, especially for homebound/bedridden patients.

I. Dental Caries

It is established that incidence of coronal caries in the old patients is more or less similar to the young; whereas, the incidence of root caries is much higher (around 40–70%). It has been recognized that in the elderly, proportion of secondary caries predominates over primary caries.

It is imperative to evaluate the risk factors involved in planning treatment of caries for the elderly. A thorough history of medical conditions and medication has to be analyzed. Treatment can be instituted in two phases:

a. Restorative phase
b. Maintenance phase

a. *Restorative Phase*

The following features are important during restoration of teeth in elderly:

i. ***Restorative considerations for coronal caries:*** In older adults, indirect restorations are generally not preferred. Direct restorations can be readily modified, repaired or replaced as incidence of secondary caries is high in such patients. Owing to high incidence of secondary caries, frequent maintenance of these restorations might be required, which is difficult in indirect restorations.

ii. ***Restorative considerations for root caries:*** Root carious lesions are predominantly seen on the proximal surfaces, followed by the facial surface. The rapid progression coupled with the thinness of tooth structure on the root may lead to pulp exposure. Therefore, even a shallow lesion should be considered potentially deep. Billings, et al (1984) had categorized root caries into four grades and outlined the treatment plan accordingly (Table 27.3).

Glass-ionomer cement is the preferred restorative material, as compared to composites (Fig. 27.4A,B). Alternative caries management strategies have also been suggested. Ozone has been used to inhibit root caries. Exposure of the lesion to ozone for 10–40 seconds is effective. Lasers have also been tried, especially for those who cannot tolerate local anesthetics; however, location of root caries may cause difficulties for the accomplishment of the procedure.

b. *Maintenance Phase*

All possible preventive measures need to be undertaken to maintain restored and the unrestored teeth.

The preventive strategy remains the same with minor modifications to suit the elderly. Daily use of fluoride dentifrices and fluoride rinses along with periodic topical fluoride application regime is advisable. Fluoride varnishes may be preferred over

Table 27.3: Grade, description and management of root caries

Grade	*Description*	*Management*
Grade I	White or light brown, surface cannot be penetrated	Topical fluorides Remineralizing agents, frequent recall
Grade II	Light brown, 0.5–1.0 mm penetration	Excavation of lesion, reshaping of margins, fluoride application
Grade III	Dark brown Penetration >1.0 mm Not extending to pulp	Restoration with glass-ionomer cement
Grade IV	Brown black Penetration into dental pulp	Root canal treatment or extraction



Fig. 27.4: Management of root caries: **(A)** Preoperative; **(B)** Postoperative

other forms. Chlorhexidine gel/mouth rinses/varnishes are advised. Every measure should be taken to prevent loss of gingival tissue attachment, which is crucial to prevent root caries.

Most elderly suffer from some degree of xerostomia. If it is drug induced, modifying the medication in consultation with the physician can be helpful. For example, substituting the drug having lesser anticholinergic effect or altering the time of medication (Salivary flow is less in the night, therefore, taking medication in the morning may improve oral comfort). The use of room humidifiers in dry weather may improve patient comfort. Instruct the patient not to use sugar-containing lozenges and candies; instead replace them with xylitol containing gums and candies. In patients with severe salivary gland dysfunction, artificial saliva can be prescribed for relief. Saliva substitutes are available in the form of gels, sprays and liquids for placement around dentures as well as on teeth and oral soft tissues.

II. Non-Carious Tooth Tissue Loss

Tooth wear is one of the most common problems in the older adults (Attrition, abrasion and erosion may lead to tooth wear). One important contributing factor is medication-induced xerostomia, which results in loss of buffering action of saliva, making the teeth more susceptible to acid erosion. With increasing age, gingival recession exposes the cementum, making it more susceptible to abrasion and erosion. Improper use of tooth brush and tooth pick/floss, etc. can also abrade the cemental surface.

Consequences of tooth wear include unsightly appearance, possible development of caries in the exposed cemental/dentinal surfaces, sensitivity and reduction in clinical crown height. Reduction in the height of the crowns is usually accompanied by progressive eruption, so that teeth continue to migrate incisally or occlusally.

The treatment for tooth wear is directed towards eliminating etiological factors and implementing the restorative measures. The treatment may be 'passive' or 'active'.

Passive treatment: It consists of monitoring and prevention. Periodic monitoring using study casts, photographs, etc. helps in assessing the progress. Preventive treatment includes dietary modifications along with fluoride regimen. Most patients can be successfully managed only by passive treatment.

Active treatment: It may be required in case of sensitivity, esthetics, functional difficulty, etc. Localized defects due to attrition, abrasion or erosion can be restored by using composite resin or glass-ionomers, depending on the location, occlusal load and esthetic needs. Very few patients need advanced rehabilitative therapy (Fig. 27.5A, B).

III. Treatment of Sensitivity

In case of sensitivity, potassium-containing desensitizing toothpastes are preferred for home use; while sodium and stannous fluorides are preferred in office treatment (Fig. 27.6). The dentin bonding agents may be effective in reducing sensitivity and preventing further damage. Root canal treatment may be carried out as the last resort for extreme sensitivity that cannot be treated by conservative methods.



Fig. 27.5: Management of non-carious defects: **(A)** Preoperative; **(B)** Postoperative



Fig. 27.6: Desensitizing agents

IV. Esthetic Rehabilitation

Smile is one of the most important attributes of human beings and is carried throughout life. Older individuals should be encouraged to improve upon their appearance, which minimizes psychological stress.

The teeth darken with age can be because of structural changes in the enamel, the addition of organic material to the enamel from the environment and due to deepening of dentinal color seen through the progressive thinning of translucent enamel. Pigmented foods consumed over a period of years may even darken teeth. The esthetic treatment for elderly includes bleaching, laminates and crowns. Any esthetic rehabilitation should be undertaken only after proper occlusal analysis.

V. Chipping and Cracking of Tooth Structure

Chipping and cracking of tooth structure is a common phenomenon in the elderly. These areas take up stains easily and appear unsightly. Occlusal and incisal surfaces (enamel into dentin) may get worn off. The dentinal cracks are generally painful because of their access to the pulp. In such cases, posterior teeth may require onlays and crowns to prevent further crack propagation. In anterior teeth, direct composites can be preferred, being less invasive.

VI. Candidiasis/Stomatitis

Candidiasis and denture stomatitis are common findings in elderly. Fissures at the corners of mouth (angular cheilitis) and petechial hemorrhage around denture-bearing areas are prevalent in older adults. Improving oral hygiene measures, low concentration of mouthwashes and antifungal therapy are the measures for management.

VII. Exposed Roots

Root exposure is universally prevalent in older individuals. The root exposure may lead to root caries and/or loss of cementum, leading to exposure of dentin. It may involve pulp. Gingival recession is defined as the apical shift of marginal gingiva from its normal position on the crown of the tooth to the level on the root beyond the cementum–enamel junction. The esthetics is also compromised in case of recession and/or root caries (Fig. 27.7A,B).

The exposed root(s) should be covered to overcome the problem of sensitivity and to improve upon the esthetics. The following conditions are mandatory for successful root coverage:

- Sufficient interdental papilla adjacent to gingival recession area
- Sufficient blood supply
- Decay/abrasion of root surface be managed prior to covering.

(Minimum of 1.0–2.0 mm attached gingiva is adequate for covering; inflammation be kept under control.)

The root coverage can be managed either by surgical or non-surgical methods.

a. *Surgical methods*

- Pedicle soft tissue graft
- Free soft tissue graft
- Double papillary flaps
- Coronally advanced flaps
- Connective tissue grafts
- Guided tissue regeneration procedures

b. *Non-surgical methods*

- Pink glass-ionomer cement (Fig. 27.8A,B)
- Gum veneers (Fig. 27.9A,B)



Fig. 27.7: Recession and caries: **(A)** Generalized recession; **(B)** Caries in multiple teeth

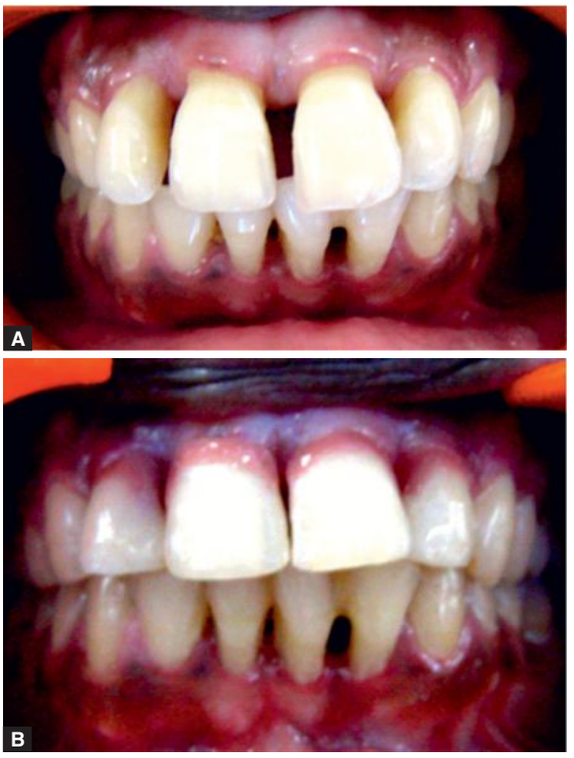


Fig. 27.8: Gingival recession: **(A)** Preoperative; **(B)** Recession restored with pink glass-ionomer cement

The non-surgical procedures are preferred in older adults as surgical procedures might not be very successful. The non-surgical methods improve esthetics, relatively inexpensive, painless and easily maintained by the older patients. They can also be used as interim measures till final treatment is being planned.



Fig. 27.9: Gum veneers: **(A)** Recession of anterior teeth (pre-operative), **(B)** Gum veneers masking the recession (post-operative)

The clinical presentation and treatment options of common oral conditions in geriatric individuals are summarized in Table 27.4.

CONSIDERATIONS FOR PRESCRIBING MEDICINES TO THE ELDERLY

The phenomena of adverse drug reactions are more common amongst elderly. The operator should know the status of prescribed drugs to the elderly; minimum medication is advised. The elderly are more sensitive

Table 27.4: Common oral conditions in geriatric individuals

Condition	*Clinical presentation*	*Treatment*
Dental caries	Brownish discoloration with or without cavitation; may not be painful	Restore with glass-ionomer; fluoride application for root caries
Non-carious cervical lesions	Attrition, abrasion, erosion and abfraction	Eliminate etiological factors; restore with glass-ionomer/composites
Periodontitis: exposed roots (gum problems)	Gingiva recession, loose teeth, swollen gums, halitosis	Scaling and root planing; adjunct antibiotic therapy, root coverage with pink composition and gum veneers
Xerostomia	Burning sensations, difficulty with speech and swallowing	Sugar-free gum/drops to stimulate salivary production; artificial saliva in severe cases
Candidiasis	Acute pseudomembranous plaques or erythematous; scaling fissures at the corners of the mouth (angular cheilitis)	Topical and systemic antifungal therapy, adequate oral hygiene/mouthwash of low concentration
Denture stomatitis	Varying erythema; along with petechial hemorrhage localized to the denture-bearing areas	Frequent removal of dentures along with antifungal therapy

to the depressant effects of drugs; hence, the dosage of analgesics, anti-anxiety drugs, sedatives will need to be reduced. A decline in the renal function occurs in the elderly and requires that dosages of drugs whose principal route of elimination is the kidney, be reduced. There is possibility of forgetfulness and poor compliance in such patients. Special packaging, clear labeling and simplified dose regimens may improve compliance.

Due diligence and interdisciplinary co-ordination among the dental and the medical professionals is crucial in providing safe, effective and appropriate care to the elderly.

The dental professionals should be conscious of the needs of the elderly patient, comprehend their psychological needs and offer them sympathetic behavior to ensure satisfaction, augment graceful dignity and inculcate a sense of well-being amongst them.

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